

Assignment: Clustering (Hands-on Exercise)

Please create a synthetic dataset and implement a clustering model end-to-end. The objective is to clearly understand each step and work with realistic data challenges.

Dataset Requirements

Your dataset should include:

- Numerical feature(s) suitable for clustering
- At least one categorical variable
- Missing values (introduced intentionally)
- A few outliers

Steps to Follow

1. Data Creation

- Generate a synthetic dataset based on a real-world scenario (e.g., segmenting customers based on purchasing behaviour, grouping patients based on health metrics, or categorizing employees based on performance data).

2. Exploratory Data Analysis (EDA)

- Visualize the data.
- Identify missing values, outliers, and relationships between variables.
- Note: Understand what to do when variables are on different scales and implement the necessary changes.
- Check for any natural groupings or patterns visible in the data.

3. Data Preprocessing

- Handle missing values.
- Encode categorical variables where necessary.
- Treat or justify the handling of outliers.
- Scale numerical features — this is critical for clustering as distance-based algorithms are sensitive to scale.

4. Model Building

- Implement a K-Means clustering model step by step.

- Understand how to determine the optimal number of clusters using the Elbow Method and Silhouette Score.
- Visualize the resulting clusters.

5. Model Evaluation

- Evaluate the clustering results using appropriate metrics (Silhouette Score, Inertia).
- Interpret what each cluster represents in the context of the real-world scenario.
- Understand why clustering evaluation is different from classification evaluation — there is no ground truth to compare against.

6. Key Observations

- Summarize learnings from each step of the process.
- Reflect on how changing the number of clusters affects the results.

Expectation

Focus on understanding and explanation at every stage.

Note: Also, try Hierarchical Clustering on the same dataset and compare the results with K-Means.